PRODUCT, ENGINEERING & OPERATIONS BLUEPRINT

On-Demand Car Wash Platform

Improved implementation plan | Version 2.0 | Decision draft

> **Recommended first launch:** prove reliable mobile doorstep washes in one city and a small number of dense service zones. Start with scheduled slots plus a limited on-demand window. Add outlet booking only when a signed anchor outlet needs it. Validate demand, partner supply, water rules, insurance and payment settlement before committing to the full build.

Executive direction

Build a dependable local service marketplace, not a broad multi-city platform on day one. The first release should make the full job loop work: quote, book, match, complete with evidence, charge, pay the partner and resolve exceptions. Keep dispatch and support operable by people while the first zones are still learning.

Planning assumptions to validate in discovery

Decision area	Recommended starting assumption
Launch footprint	One city; two or three adjacent, dense zones with clear service boundaries. Expand only after supply and demand are balanced.
Service model	Mobile doorstep wash first. Keep outlet inventory and POS as a later module unless an anchor outlet is already committed.
Booking promise	Scheduled slots are the reliable baseline. Offer on-demand only during staffed hours and within measured travel-time limits.
Partner model	Verified independent partners with explicit operating zones, hours, service skills, equipment checks and payout terms. Confirm local worker classification and tax treatment.
Payments	Select a locally supported payment provider that supports the intended marketplace and partner payout model. Keep provider integration replaceable.
Plan horizon	Use a 22-24 week planning envelope for a staffed MVP and closed pilot; re-estimate after discovery, provider onboarding and field trials.

Decisions required before engineering is locked

- Launch geography, currency, service area, languages and supported payment methods.
- Which wash methods are allowed at homes, apartments, basements and commercial sites; who supplies water and power?
- Partner verification, insurance limits, tax, refund policy, customer damage claims and settlement schedule.
- Which service packages and add-ons will be offered, and what promise can operations actually meet?

WHAT SHIPS FIRST

01. Product scope and service journeys

The original four-portal concept is a useful destination, but it is too much to validate as one MVP. Ship customer and partner apps with a lean web operations console. Add outlet tools and advanced growth features after the local service loop is stable.

Release boundary

MVP: required for closed pilot	Later: add when evidence supports it
Customer account and vehicle garage; address pin plus access notes; service catalog and transparent quote; scheduled booking and capped on-demand availability.	Outlet discovery, bay calendars, walk-in POS, multi-tenant outlet inventory and outlet-specific staff roles.
Partner availability, service skills, offer accept/decline, navigation handoff, arrival confirmation, guided before/after evidence and earnings summary.	Automated subscriptions, family/fleet accounts, advanced route batching, partner lending or equipment financing.
Operations console for partner approval, zones, service windows, live job exceptions, manual assignment, refunds and support cases.	Predictive staffing, complex surge pricing, loyalty tiers, broad referral automation and multi-city tenancy.
Push and in-app status updates; masked call/chat for active jobs; payment authorization, capture, refunds and payout reconciliation.	Automated quality scoring from images, real-time location sharing outside active work, AI damage judgments.

Customer journey

- **Choose:** add a vehicle, select a supported service and location, provide parking/access details, choose a time window and review the itemized total.
- **Confirm:** accept the quote and cancellation terms; authorize payment; receive a realistic arrival window only after partner assignment.
- **Complete:** see job status, communicate through masked channels, review pre-wash condition photos, then receive completion evidence and an easy issue/report flow.

Partner journey

- Set shift, service radius and capabilities; receive a time-limited offer with payout estimate, vehicle/service details and access notes; accept or decline.
- Navigate, mark arrived, confirm the customer or booking code, capture guided condition evidence, complete checklist, upload after-photos and close the job.

Booking lifecycle and exception states

> **Requested > Quote accepted > Dispatching > Assigned > En route > Arrived > In service > Completed > Payment captured > Partner payout eligible.** Every transition is timestamped and attributed. Add explicit branches for expired offer, customer cancellation, partner no-show, unsafe site, payment failure, disputed work, refund and reassignment. An operator can correct a job only with a reason recorded in the audit history.

BUILD FOR LEARNING, SPLIT ONLY WHEN NEEDED

02. Architecture and technology

Use managed infrastructure and a modular backend first. The original plan names Kubernetes and multiple microservices before traffic or team boundaries are known; that adds operating work without improving the first pilot. Keep modules clean so a proven high-load component can be extracted later.

Reference architecture

Layer	Starting choice and responsibility
Mobile	Flutter is a reasonable shared-code option if a short device spike proves reliable background location, camera uploads, permissions and battery use on supported phones. Keep customer and partner experiences distinct.
Web	React-based responsive operations console; role-based access and audit trail for sensitive actions.
Application	TypeScript backend (NestJS or equivalent) as a modular monolith: identity, catalog, quotes, booking, dispatch, partner, payment ledger, notification and support modules. Versioned REST APIs; WebSockets for active job updates.
Data	PostgreSQL with PostGIS for bookings, zones and spatial matching. Redis only for short-lived presence, offer locks and rate limits; it is not the source of truth.
Async work	Managed queue plus transactional outbox for notifications, image processing, payment webhooks, payout reconciliation and retryable background work.
Files and providers	Private object storage for evidence; push notification provider; maps/geocoding/road ETA provider; local marketplace-capable payment provider; masked communications provider. Abstract provider-specific APIs.
Runtime	Managed application hosting, managed database, secrets manager, infrastructure-as-code, separate dev/staging/production accounts and central metrics/logging. Add Kubernetes only when workload, compliance or deployment needs justify it.

Real-time and resilience rules

- Send frequent location updates only while a partner is on shift or serving an active job; use state-based intervals and stop collection when off duty.
- Use WebSockets for active trip updates with push or polling fallback when connectivity drops. Show the last update time and never present stale GPS as live.
- Make booking, offer acceptance, payment callbacks and completion idempotent. Use bounded retries, dead-letter handling and operator visibility for failed jobs.
- Set backup retention, recovery objectives and a restore drill before the pilot. Alert on booking/payment failures and queue delay, not just server uptime.

> **Scale trigger:** split a module only when measured load, independent release cadence, reliability isolation or team ownership makes that separation worthwhile. Keep PostgreSQL transactions and a clear event log authoritative during the pilot.

PROTECT THE SERVICE PROMISE

03. Booking, dispatch, pricing and payments

Dispatch logic

Step	Rule
1. Eligibility	Filter by zone polygon, open hours, service type, partner status, skills/equipment, capacity, travel limit and any site restrictions.
2. Candidate ranking	Rank by road ETA and predicted arrival reliability, then apply a modest fairness factor for workload and recent offer history. Straight-line distance alone is not enough.
3. Offer and lock	Send time-limited offers to one or a small batch of eligible partners. First valid acceptance wins an atomic booking lock. Expire, retry and widen the search within configured caps.
4. Promise and monitor	Only show a firm arrival window after acceptance. Detect stale location and missed milestones; alert operations before automatic reassignment.
5. Recovery	Provide manual assignment, customer notification, cancellation/refund path and partner unavailability controls. Reassignment must not create duplicate partner work or payment capture.

Pricing rules

- Start with a transparent quote: service base + vehicle tier + selected add-ons + zone/travel component + taxes/fees where applicable - discounts. Store each line and the ruleset version.
- Show the full total, expected duration, included/excluded tasks and quote expiry before authorization. Lock the accepted quote; any change requires customer approval.
- Avoid weather and demand surcharges at launch. If later introduced, cap them, display the reason and amount before booking, and monitor conversion and partner earnings.

Payments and settlement

- Authorize at booking only when the scheduled start falls inside the provider authorization window. For longer lead times, either collect a clearly disclosed booking charge or re-authorize near service; capture at completion according to provider capabilities and local rules. Never store raw card credentials; use provider tokens.
- Maintain an internal ledger for gross charge, tax, platform fee, partner payable, refund, promo and payout reference. Reconcile provider transactions and bank/payout files daily.
- Make refunds, partial refunds, chargebacks, failed payout and cancellation fees explicit states. Record who approved every manual adjustment.
- Do not launch subscriptions until pauses, skipped washes, failed renewal, cancellation, unused balance, partner capacity and refund handling are defined.

Customer communication and safety

Mask phone numbers and keep chat/calls available only for the active booking plus a short support window. Give both sides a fast unsafe-site, damage and emergency report action. Keep customer-facing ETAs honest and notify on assignment, delay, arrival, start, completion and cancellation.

MAKE EVERY JOB AUDITABLE

04. Data model, APIs and integration controls

Use relational records for current business truth and append-only events for history. Normalize operational entities; store image evidence in private object storage with database references and retention metadata, not as large database blobs.

Core entities

Domain	Minimum records and important fields
Identity	User, role grants, partner profile, verification status, consent and notification preferences. Keep identity verification documents separate from general support access.
Supply and catalog	Vehicle, service, add-on, price rule version, service zone polygon, partner service skill, equipment checklist, shift and capacity.
Transaction	Quote, booking, booking event, dispatch offer, cancellation, support case and customer/partner acknowledgement. Include timestamps, currency and actor.
Evidence	Job evidence record with type, uploader, capture time, upload status, object key, checksum, access policy and retention deadline.
Money	Payment attempt, immutable ledger entry, refund, partner payable, payout batch and provider reconciliation item. Store amounts in integer minor units plus currency.
Later	Subscription, outlet, bay, inventory lot, corporate account and referral attribution only when those products are approved.

API and event design

Interface	Minimum controls
Booking commands	POST quote, POST booking, accept/decline offer, mark arrived, start, upload evidence and complete. Enforce role and ownership at every endpoint.
Partner location	Authenticated update with timestamp, accuracy and active-shift status. Reject stale/out-of-order updates and minimize retention.
Provider webhooks	Verify signatures, deduplicate provider event IDs, persist before processing, retry safely and surface unmatched events for reconciliation.
API safety	Idempotency keys for create/capture/refund actions; versioned schemas; explicit error codes; request IDs; rate limits; audit log for admin overrides.

Operational data decisions

- Store all timestamps in UTC and render in the customer/partner local time zone. Keep currency explicit and amounts in minor units.
- Use a booking state machine enforced server-side; a client must not be able to jump from assigned to paid or alter price after acceptance.
- Use the transactional outbox so a database commit cannot succeed while losing the dispatch/notification event. Track event delivery status.
- Document evidence retention, deletion requests, legal holds and access logs. Do not label object storage immutable unless the selected configuration and retention policy enforce that property.

THE PHYSICAL SERVICE IS PART OF THE PRODUCT

05. Operations, quality, safety and compliance

Partner readiness

- Verify identity, payout account, required business documents and eligibility checks for the launch jurisdiction; record expiry dates and renewal owners.
- Train on each offered package, customer communication, safe chemical handling, photo evidence, vehicle access, site permissions and incident escalation.
- Issue or inspect a standardized kit: approved cleaners with current safety data sheets, separated/clean towels, PPE, safe power/water equipment, waste containment where required and a replenishment checklist.
- Use a short observed practical assessment before activation. Suspend new offers when certification, required kit or insurance status expires.

Quality and damage handling

- Capture guided pre-service condition photos before work begins and after-service photos before the job closes. Make lighting, angle and vehicle identity checks part of the partner flow.
- Ask the customer to confirm the vehicle and pre-existing condition when present; allow a no-contact flow with clear proof and a support route.
- Define a human-led dispute process: evidence freeze, response deadlines, insurer/partner escalation, refund authority and customer updates. Image models may help triage later but must not decide liability.
- Sample completed jobs for quality review, coach repeat defects and pause partners after serious safety or damage incidents pending review.

Water, site and environmental rules

> Before launch, map which methods are permitted at each site type and confirm local water-use, runoff, chemical storage/disposal, building access and noise requirements with qualified local advisers and property operators. Avoid a blanket water-saving percentage or compliance claim unless the process has been measured and substantiated. If a site cannot be served safely, the app must block that service or offer another location/method.

Privacy, security and support

- Collect precise location only during declared partner availability and active work; explain it in onboarding and provide a visible off-duty state.
- Use least-privilege roles, MFA for staff, encrypted transport/storage, managed secrets, admin access logs and a documented breach/incident contact tree.
- Provide staffed pilot support during all bookable hours, with a live exception queue for unassigned, late, failed-payment, unsafe-site and damage cases.
- Confirm privacy notices, consumer terms, partner contracts, insurance scope, tax treatment, payments/marketplace obligations and app-store disclosures with local professionals.

22-24 WEEK PLANNING ENVELOPE

06. Delivery roadmap and phase gates

This is a planning estimate, not a fixed commitment. Run customer/partner validation, legal review and provider feasibility in parallel with design. Re-estimate after the selected payment provider and service model are proven in sandbox and in the field.

Phase	Weeks	Outputs and exit gate
0. Validate	1-2	Interview customers, washers and property managers; map site/service rules; confirm unit economics inputs, partner supply and payment provider marketplace/payout support. Gate: one viable zone and a written operating model.
1. Foundation	1-4	Clickable prototype, service blueprint, booking state machine, zone model, data/privacy plan, vendor choices, deployment pipeline and support playbook draft. Gate: product/ops sign-off and technical spikes for location, camera upload and payout.
2. MVP build	5-12	Customer + partner apps, admin console, catalog/quotes, booking, manual-assisted dispatch, payments sandbox, evidence upload, masked comms and basic reporting. Gate: end-to-end happy and failure flows work in staging.
3. Internal alpha	13-15	Staff/partner dry runs; device and network checks; payment/refund/payout reconciliation; partner training; security and restore review; operational dashboards. Gate: no unresolved critical safety, money or privacy defects.
4. Closed pilot	16-20	Two or three bounded zones, staffed support, controlled partner cohort and capped booking volume. Daily review of fulfilment, ETA, quality, partner earnings and contribution margin. Gate: meet agreed pilot thresholds for two consecutive weeks.
5. Soft launch	21-24	Open the proven zones gradually; marketing and promotions remain capped; publish support hours and service exclusions. Expand one zone at a time after supply and economics hold.

Planning team

Role	Indicative allocation
Product manager + operations lead	Own scope, field decisions, partner supply, policy and pilot gates.
Product designer	One lead through research, service blueprint, prototype, field usability and final content.
Mobile engineering	Two engineers for customer and partner flows, camera/location behavior and release readiness.
Backend engineering	Two engineers for booking, dispatch, payment ledger, APIs, data model and integrations.
QA / test engineering	One engineer focused on booking/payment state coverage, device/network behavior and pilot regression.
Platform/security + support	Fractional platform/security ownership; named support lead and trained pilot agents before bookings open.

SCALE ONLY WHAT WORKS PER ZONE

07. Measurement and unit economics

Use completed, on-time washes with positive contribution margin as the operating north star. Report by zone, service package, time window and partner cohort; overall averages can conceal an unworkable zone.

Pilot dashboard

Measure	Definition	Initial pilot target / use
Fulfilment	Completed jobs / confirmed bookings, excluding customer cancellations before dispatch.	At least 90% after the first stabilization week; investigate every miss.
On-time arrival	Partner arrives within the promised window / assigned jobs.	At least 85% in pilot; tune promise windows using measured road ETA.
Customer rating	Average completed-job rating with response count and distribution.	At least 4.6/5 once there are 100+ responses; review low-score reasons.
Rework / refund	Jobs with a quality rework or service-related refund / completed jobs.	Below 5% as an initial hypothesis; inspect reason codes weekly.
Partner utilization	Booked service time / declared available time, by zone and shift.	Track balance; avoid scaling marketing when partners are already overloaded.
Contribution margin	Net revenue less partner payout, payment fees, supplies, refunds/rework, variable support and promotions.	Positive by zone before expansion; include one-time kit subsidies separately.
Repeat and subscription	Repeat booking within 30 days; subscription renewal and pause/cancel reasons.	Establish baseline during pilot. Do not optimize subscriptions before wash quality and capacity stabilize.
Safety and trust	Serious incidents, damage claims, unresolved disputes and location/privacy complaints.	Zero unresolved high-severity incidents at any launch gate.

Unit economics worksheet

> **Contribution per completed wash** = customer cash collected (excluding pass-through tax) - partner payable - payment fee - consumables/water - variable support - expected refunds/rework - promotional subsidy. Track customer acquisition cost and partner acquisition/equipment costs separately. Use actual pilot data, not forecast conversion alone, to decide whether a zone can expand.

Review daily during pilot: booking funnel, unassigned jobs, median and P90 ETA, cancellations by reason, wash duration, photo exceptions, payment/payout exceptions, supply coverage, contribution per job and cohort repeat. Set targets with the operations lead after two weeks of baseline data; the numbers above are starting hypotheses.

MAKE THE LAUNCH REVERSIBLE

08. Risk register and go-live checklist

Risk	Early control and response
Insufficient local density	Constrain bookable zones/hours, run scheduled-first, cap promotions and use manual dispatch. Do not advertise a service window without staffed supply.
Damage or inconsistent quality	Guided condition evidence, trained partners, sampling, transparent claims workflow, insurer coordination and pause authority.
Water/site restrictions	Site-by-site method matrix; block unsupported methods; document property approval and runoff/waste handling before activation.
No-show or late partner	Offer expiry, stale-location signal, customer notice, manual fallback, partner coaching and no-penalty reassignment for customer impact.
Payment or payout mismatch	Provider sandbox certification, idempotent webhooks, daily ledger reconciliation, manual exception queue and refund approval limits.
Location/privacy complaint	Explicit on-shift controls, visible status, minimal retention, role-based access and documented deletion/support path.
Demand spike or weather shock	Bookable capacity caps, configurable pause controls, customer-visible limits and incident messaging. Avoid unvalidated surge behavior.

Go-live checklist

- Market, property access, water/runoff, chemical handling, tax, worker classification, consumer policy, insurance and privacy reviewed for the selected geography.
- Payment authorization, capture, cancellation, refund, chargeback, partner payout and end-of-day reconciliation proven in provider sandbox and controlled live transaction.
- Service catalog, inclusions/exclusions, price, quote expiry, time windows, service-zone rules and customer notices approved by operations.
- Partner checks, practical training, kit checklist, evidence guide, safe-site policy and incident contact are active.
- Support staffing and escalation path cover every bookable hour; operator can pause zones, pause a partner and manually recover a booking.
- Monitoring covers job lifecycle, payment exceptions, queue delays, location freshness, crashes, upload failures, backups and restore.
- Privacy disclosure covers precise location and photos; retention schedule, access permissions and deletion/contact process are documented.
- Pilot exit thresholds are signed by product, engineering and operations; expansion is limited to one additional zone at a time.

> **Recommended launch gate:** run the closed pilot with a capped customer cohort. Expand only after fulfilment, on-time arrival, service quality, unit economics and safety thresholds hold for two consecutive weeks and unresolved high-severity incidents are zero.

This plan is a product and delivery blueprint. Local legal, insurance, environmental, employment and payment requirements must be confirmed for the actual launch market before public bookings begin.